Assignment 2: Pricing and Hedging Climate Risks

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Deadline: November 18, 2026, 23:59

General information

Submission. Submit your results and answers to the assignment's questions in a report PDF. Additionally, you will have to submit the code you used to generate these results. For the data analysis, please use Python. Do not submit a zip file. Instead, upload the PDF and the Python file separately. Do not upload any data files. Name all files: *Assignment2_Group#*, adding the respective file ending. **Not following these instructions will result in a one grade point deduction.**

Grading. The assignment will be worth 1/6 of your final grade. The assignment will be graded based on three components:

- *Functioning of the code.* The sole requirement is that the code runs without errors on our machines and produces the results in your report. Include all necessary libraries/modules, give instructions on packages to install if necessary, and indicate path variables that must be changed when running the code on another machine.
- *Quality of the analysis.* We will evaluate the quality of the analysis based on the correctness and clarity of the results, e.g., whether the figures and regression coefficients are correct and the appropriateness of the methods used.
- *Clarity of the presentation.* We will evaluate the clarity of the presentation of the results. Describe your results clearly and exhaustively, consistent with potential visualizations and calculations. Comprehensively label all figures and tables. To answer the questions, do not use bullet points. Think as if you were writing a report for a client or a research paper.

Assignment

Data. The main dataset for this assignment is the file "assignment_data.xlsx" provided in the assignment folder. The file contains quarterly stock prices from 2009 Q4 to 2019 Q4 combined with the climate risk measure from Sautner et al. (2023) and risk factors from Kenneth French's data library.

The file "assignment_data.xlsx" contains (among others) the following variables:

- *ISIN*: The International Securities Identification Number of the stock.
- *Date*: The quarter of the stock return expressed as the last date of a quarter.
- *Price Close*: The stock's closing price at the end of a quarter.
- *std_risk*: The regulatory climate change risk measure from Sautner et al. (2023), standardized cross-sectionally in each quarter. (Higher values indicate a higher transition risk exposure.)
- *Mkt-RF*: The excess return of the market portfolio in the quarter.
- *SMB*: The size factor in the quarter.
- *HML*: The value factor in the quarter.
- *RF*: The risk-free rate in the quarter.

Additionally, there is a dataset containing the data on the WSJ index from Engle et al. (2020).

1 Risk and Returns [45 points]

(a) For each quarter, form equal-weighted quintile portfolios sorted on the climate risk exposure measure *std_risk* of the previous quarter. Plot the cumulative portfolio excess returns for the top quintile and bottom quintile portfolios over time. Do the results support the existence of a risk premium for "brown" assets? Justify your answer (max. 5 sentences). **[10 points]**

(b) Plot the cumulative excess returns of a Green-Minus-Brown (GMB) portfolio over time. Interpret the results. **[10 points]**

(c) Referring to the model of Pastor et al. (2021), state two potential explanations for the previous results. (No calculations needed.) **[10 points]**

(d) Plot the cumulative excess returns of the GMB portfolio over time and add to the plot the cumulative sum of WSJ index innovations (Engle et al., 2020). Interpret the results. (*Hint:* Take the average of the monthly innovations within a quarter before cumulating them.) **[5 points]**

(e) Can we expect green stocks to outperform brown stocks in the future? Justify your answer (max. 5 sentences). (No calculations or figures are needed.) **[10 points]**

2 Hedging Transition Risk [15 points]

(a) State the objective for hedging transition risk in a stock portfolio. **[5 points]**

(b) How well would the GMB portfolio have hedged (future) innovations in the WSJ index? Compare the GMB portfolio to the market portfolio, the size factor, and the value factor. **[10 points]**

References

Engle, R. F., Giglio, S., Kelly, B., Lee, H., and Stroebel, J. (2020). Hedging Climate Change News. *The Review of Financial Studies*, 33(3):1184–1216.

Pastor, L., Stambaugh, R. F., and Taylor, L. A. (2021). Sustainable investing in equilibrium. *Journal of Financial Economics*, 142(2):550–571.

Sautner, Z., Van Lent, L., Vilkov, G., and Zhang, R. (2023). Firm-Level Climate Change Exposure. *The Journal of Finance*, 78(3):1449–1498.